PackEats

Section 2
Group 7
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Mission Statement

PackEats is a secure and sustainable food delivery system designed for NC State University. It addresses the logistical and environmental challenges of high-volume on-campus dining by integrating smart recommendations, eco-friendly delivery options, and optimized order management. By leveraging role-based access, modular architecture, and real-time synchronization, PackEats supports restaurants, students, and administrators in a single scalable ecosystem.

Key stakeholders

Students (Customers & Eco-Drivers)	Restaurants	Communities	Sustainability Partners
Students benefit from affordable meal options and the opportunity to earn through flexible part-time delivery roles, promoting both convenience and financial independence	Restaurants gain a streamlined digital system for managing menus, receiving orders, and expanding their customer base within the university ecosystem.	Residents enjoy accessible, low-cost, and sustainable food delivery services that promote environmental responsibility and local engagement.	Organizations and initiatives promoting eco-friendly transportation collaborate to expand green delivery models and reduce the carbon footprint of urban logistics.

System Capabilities (v1.0)

- Secure Multi-User Authentication: Separate login and registration for customers, restaurants, and drivers. Encrypted credentials, role-based access, and full database.
- Dynamic Restaurant & Menu Browsing: Live restaurant and dish data pulled directly from PostgreSQL. Personality-based quiz recommends dishes and syncs with user cart in real time.
- Seamless Checkout & Order Management: Integrated payment options (card, meal plan, eco-driver selection) with atomic transactional updates in the backend.
- Restaurant Dashboard: Real-time order and menu management system with update controls and order tracking.
- Driver Dashboard: A dedicated panel for delivery partners to view and accept orders

Testing and Maintainability

179 automated test cases.

- **127** frontend → **92%** line coverage
- **52** backend → **87%** coverage

Continuous Integration pipeline ensures all commits are verified before deployment.

Upcoming Features (v1.2)

- Enhance Driver Dashboard: Upgrade the driver interface to provide better visibility of pending orders, filter by Eco or Standard deliveries, show estimated delivery times, and include real-time location tracking synced continuously to the database.
- Driver Performance & Incentives: Monitors driver activity through delivery completion rates, feedback, and eco-contributions. High performers are recognized on leaderboards and rewarded through bonuses or milestone-based incentives.
- Sustainability Tracker: Tracks the environmental benefits of eco-deliveries by measuring emissions saved and green delivery counts. It promotes awareness and encourages both drivers and customers to choose sustainable options.
- Smart Restaurant Promotions: Give restaurants the ability to launch dynamic discounts or flash sales during low-demand hours or when managing surplus food.

Technology Stack

- Frontend: React.js + HTML + CSS
- Backend: Java Spring
- Database: PostgreSQL
- Testing: JUnit, Jest
- CI/CD: GitHub Actions

Discussion Forum/Links

Discord chat: discord

Github repo: CSC-section 2- group 7

Website demo: demo video